

## 24. Window ki Size aur Scrolling

Root element `document.documentElement` (`<html>`) hai.

### Window ki width/height

```
js

document.documentElement.clientWidth
document.documentElement.clientHeight
```

Ye **scrollbar ko hata ke** available (visible) area deta hai. Ye hi usually chahiye hota hai.

`window.innerWidth/innerHeight` **scrollbar ko include** karte hain, to dono alag values de sakte hain.

Hamesha `<!DOCTYPE HTML>` likho, warna geometry properties ajeeb behave kar sakti hain.

### Poore document ki height (scroll hua hissa bhi)

`documentElement.scrollHeight` akela bharsemand nahi (kuch browsers me `clientHeight` se bhi kam aa sakta hai). Isliye **maximum** lo:

```
js

let scrollHeight = Math.max(
  document.body.scrollHeight, document.documentElement.scrollHeight,
  document.body.offsetHeight, document.documentElement.offsetHeight,
  document.body.clientHeight, document.documentElement.clientHeight
);
```

(Ye purane zamane ki inconsistencies ki wajah se hai.)

### Current scroll padhna

```
js

window.pageYOffset // upar se kitna scroll
window.pageXOffset // left se kitna
```

- Read-only. Ye `window.scrollY`/`window.scrollX` ke alias hain.

### Page ko scroll karna

Page ka DOM pura ban chuka ho tab hi kaam karta hai (`<head>` ki script me nahi chalega).

- `window.scrollBy(x, y)`: **current position se relative** (jaise `scrollBy(0, 10)` = 10px neeche)
- `window.scrollTo(pageX, pageY)`: **absolute position** (jaise `scrollTo(0, 0)` = sabse upar)
- `elem.scrollToView(top)`:
  - `top = true` (default): element window ke **upar** align
  - `top = false`: element window ke **neeche** align

## Scrolling band karna

js

```
document.body.style.overflow = "hidden"; // scroll freeze
document.body.style.overflow = "";      // wapas
```

Dusre elements par bhi kaam karta hai. Nuksaan: scrollbar gayab hone se content "jump" kar sakta hai. Fix: pehle-baad `clientWidth` compare karke `padding` daal do.

## Summary

| Kaam                   | Kaise                                                                       |
|------------------------|-----------------------------------------------------------------------------|
| Visible window size    | <code>documentElement.clientWidth/Height</code>                             |
| Poore document ki size | upar wala <code>Math.max(...)</code>                                        |
| Current scroll         | <code>window.pageYOffset / pageXOffset</code>                               |
| Scroll karna           | <code>scrollTo</code> , <code>scrollBy</code> , <code>scrollIntoView</code> |